



Rebels without a clause: Processing reflexives in fronted *wh*-predicates

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ABSTRACT

English reflexives like *herself* tend to associate with a structurally prominent local antecedent in online processing. However, past work has primarily investigated reflexives in canonical direct object positions. The present study investigates cataphoric reflexives in fronted *wh*-predicates (e.g., *The mechanic that James hired predicted how annoyed with himself the insurance agent would be*). Here, the reflexive is encountered in advance of its grammatical antecedent. We ask two questions. First, will readers engage an anaphoric (backwards-looking) or cataphoric (forwards-looking) search for an antecedent? Two, how similar is this process to the retrieval process for direct object reflexives? In two eye-tracking experiments, we found that readers initially interpret a cataphoric reflexive anaphorically, and tend to associate the reflexive with the more recently encountered antecedent. We propose that structural guidance for reflexive resolution occurs only when the necessary configurational syntactic information is available when the reflexive is encountered.

Introduction

Anaphors such as *him* or *herself* are ubiquitous in natural language. As comprehenders, we often understand them accurately and effortlessly. However, there remain many open questions concerning the cognitive mechanisms that make this possible. These questions have proven difficult to address, given the diverse processing challenges posed by anaphoric expressions. Anaphors are underspecified in their reference; a pronoun *him* or *herself* could in principle refer to any male human individual. Comprehenders must infer the intended reference in a given context. This process often relies on another, often linguistically overt expression called the *antecedent*. For example, in a sentence like *While John was reading, he ate an apple*, comprehenders must relate *he* to a preceding antecedent *John* to recover the reference. This process is often characterized as a type of memory retrieval process that links the anaphor to its antecedent (e.g., Foraker & McElree, 2007; for a review, see Sturt, 2013). However, in other linguistic contexts, the linear relationship between the anaphor and its antecedent can be reversed, creating a *cataphoric* dependency between the pronoun and its antecedent. For example, in *While he was reading, John ate an apple*, a comprehender must maintain the pronoun in memory and associate it with a subsequent noun in the sentence (Van Gompel & Liversedge,

2003; Kazanina, Lau, Lieberman, Yoshida, & Phillips, 2007). This creates a different sort of processing challenge.

In the present study we investigate the processing of cataphoric reflexive pronouns in sentences like *John wonders how proud of himself Paul was*. We will refer to reflexives as *cataphoric* because in this sentence, the reflexive anaphor *himself* linearly precedes its antecedent, *Paul*; in this sense, they may be contrasted with the more widely studied anaphoric reflexives. However, as we will see, it is not clear that comprehenders actually process these reflexives as if they were cataphoric in real-time processing. We contrast the real-time processing of these cataphoric reflexive pronouns with that of the more widely-studied anaphoric reflexives. For anaphoric reflexives, there is a general consensus that structural constraints are applied immediately in comprehension to restrict the antecedent search process for anaphoric reflexives (for a review, see Sturt, 2013). However, when a reflexive precedes its antecedent, a structurally-guided backwards retrieval process is not a reliable strategy for identifying the reflexive's antecedent. Instead, the only way to identify the reflexive's antecedent in a strictly structure-guided manner would involve a forwards search through the sentence to link the reflexive to its antecedent.

This contrast in the processing strategies necessary to ensure a grammatically appropriate interpretation for anaphoric and cataphoric

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reflexives raises two questions that we address in this paper. The first question is one of incrementality: do comprehenders strictly engage a forward search to identify the antecedent for a cataphoric reflexive, processing it in a manner similar to other cataphoric pronouns (Van Gompel & Liversedge, 2003; Kazanina et al., 2007)? Or will they engage a backwards retrieval process to identify an antecedent even in this context, attempting to form an anaphoric link with an antecedent in the preceding context? To preview our results, we find that comprehenders do engage a backwards retrieval process to link a cataphoric reflexive to a previously encountered antecedent. In other words, comprehenders treat cataphoric reflexives as if they were anaphoric in real-time processing, despite the fact that this strategy may yield an ungrammatical interpretation. This surprising finding replicates earlier results (Omaki, 2010). This finding leads to our second question: how similar is the retrieval process engaged for anaphoric reflexives and cataphoric reflexives? To preview our results, we observe a contrast between the two. We replicate the widely observed finding that anaphoric reflexives retrieve structurally prominent, c-commanding antecedents. However, in two experiments, cataphoric reflexives retrieve non-c-commanding, linearly proximate antecedents over more structurally prominent antecedents. We propose that structural information guides the processing of reflexive pronouns only when there is enough structural information available to do so.

Background on structural constraints on reflexive processing

In contrast to pronouns such as *he* or *him*, reflexives are subject to very strict constraints on their interpretation. These restrictive interpretive constraints on reflexives have made reflexives an interesting test case for theories of syntactic processing. Perhaps the most well-known formulation of these licensing conditions is Chomsky's (1981, 1986) Binding Theory. On this theory, the reference for reflexive pronouns is fixed by Principle A, which states that a reflexive's antecedent must be (i) c-commanding (a sort of structural prominence) and (ii) in the local syntactic domain. For example, in the sentence *Jorge said that the man that Loki bit embarrassed himself*, many English speakers have the intuition that only *the man* is a possible antecedent for the reflexive *himself*. According to the Binding Theory, this is because *Loki* is not structurally prominent enough in the sentence to serve as an antecedent (it does not c-command the reflexive), and *Jorge* is not structurally local enough (it is in a separate clause). These restrictive structural constraints suggest one natural question for incremental sentence comprehension: do the structural constraints expressed in Principle A of the Binding Theory apply immediately in comprehension to restrict the search for a reflexive's antecedent?

There is to date a substantial empirical literature that supports the view that the structural constraints on reflexives can indeed be applied immediately in comprehension. Nicol and Swinney (1989) were one of the first to demonstrate this using a cross-modal priming paradigm. They auditorily presented participants with sentences such as *The boxer told the skier that the doctor for the team would blame himself for the injury*. At the offset of the reflexive, participants were given a visual lexical decision task. Nicol and Swinney observed priming for lexical associates of *doctor*, but not for *boxer* or *skier*. Since *doctor* is the only structurally permitted antecedent, Nicol and Swinney concluded that only antecedents that meet the structural requirements of Binding Theory were reactivated when the reflexive was processed. In another important study, Sturt (2003) investigated the processing of reflexives using eye-tracking-while-reading. In his study, Sturt used a *mismatch paradigm* to investigate the incremental processing of reflexives. Sturt factorially manipulated whether the reflexive's gender feature matched the (stereotypical) features of a structurally accessible antecedent (e.g. *the surgeon* in (1)) or those of a structurally inaccessible antecedent (e.g. *he/she*):

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- (1) **CONTEXT:** Jonathan/Jennifer was pretty worried at the City Hospital.
TARGET: He/she remembered that the surgeon had pricked himself/herself with a needle.
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Sturt observed a disruption in early reading time on the reflexive anaphor when it mismatched the stereotypical gender feature of the accessible antecedent *the surgeon*. The features of the inaccessible antecedent did not impact reading times in early measures, although they did influence second pass reading times. From this Sturt concluded that the Binding Theory is deployed as an early, but defeasible, filter on the antecedent search process. Subsequent research has generally supported the conclusion that structural constraints on reflexives are applied early in comprehension. For the most part, Sturt's conclusion that inaccessible antecedents do not influence the processing of reflexives in early measures is borne out by a range of studies using the eye-tracking-while-reading technique (for a review and meta-analysis, see Sturt, 2013; Jäger, Engelmann, & Vasishth, 2017), as well as studies using ERP methods (Xiang, Dillon, & Phillips, 2009). However, more recent work by Parker and Phillips (2017) has shown that inaccessible antecedents can have an early influence on the processing of reflexives if the accessible subject mismatches the reflexive on more than one morphosyntactic feature (for related findings, see also Badecker & Straub, 2002; Patil, Vasishth, & Lewis, 2016; Sloggett, 2017). In addition, Kaiser, Runner, Sussman, and Tanenhaus (2009) used a visual world eye-tracking experiment and found that discourse-level factors had an early impact on the processing of reflexives as well, with reflexives preferentially taking antecedents that are sources of information, even though those antecedents did not meet the structural constraints of Binding Theory.

Processing cataphoric reflexives

On balance, the evidence to date supports the claim that the structural constraints on reflexives are used as highly-weighted cues to retrieve the antecedent of a reflexive (Cunnings & Sturt, 2014). In this sense, we may say that structural constraints are immediately applied in comprehension, guiding interpretation of a reflexive pronoun. As the brief review above makes clear, however, there is evidence that these structural cues may sometimes be overridden by other discourse or morphosyntactic constraints.

This conclusion largely rests on investigations of anaphoric reflexive pronouns in direct object position, as in (1) above. A notable exception to this is studies that have investigated the processing of reflexives in 'picture noun phrase' contexts, where the reflexives are embedded inside a larger direct object noun phrase (Cunnings & Sturt, 2014; Kaiser et al., 2009; Runner, Sussman & Tanenhaus, 2003, 2006). Even so, in all studies described above, the antecedent linearly precedes the reflexive, creating an anaphoric relationship between a reflexive and its antecedent. Under these conditions, it is natural to model the processing problem for reflexives as a memory retrieval problem, with the various constraints on reflexives providing retrieval cues that gate access to antecedent representations in memory (Cunnings & Sturt, 2014; Dillon, 2014; Jäger et al., 2017; Sturt, 2013; see also Kaiser et al., 2009).

However, reflexives need not always follow their antecedent. In some constructions, it is possible for the reflexive to precede its antecedent, creating a cataphoric dependency between the reflexive and its antecedent. Two examples of where this may occur are in *wh*-argument fronting (2) and in *wh*-predicate fronting (3):

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- (2) *wh*-argument fronting
 Alice recalled which drawing of herself the attractive young nanny had damaged ___ during the summer vacation.
- (3) *wh*-predicate fronting
 Alice recalled how pleased with herself the attractive young nanny had been ___ during the summer vacation.
-

In both (2) and (3), the underlined phrase is a sort of ‘filler’ that has been displaced from its canonical position in the sentence. In the *wh*-argument fronting example, the filler comes from the direct object position. In the *wh*-predicate fronting example, the filler is the entire predicate of the embedded clause.

This displacement has both grammatical and processing consequences. It has been observed in the syntax literature that the interpretive possibilities for the reflexive is different in (2) and (3) (Barss, 1986; Fox & Nissenbaum, 2004; Heycock, 1995; Huang, 1993; Takano, 1995; for a review, see Sportiche, 2006). In *wh*-argument fronting contexts (2), the reflexive may be interpreted as either the matrix subject *Alice*, or as the embedded subject *the nanny*. In the terms of the current paper, this reflexive is genuinely ambiguous between a cataphoric and anaphoric interpretation. However, in *wh*-predicate fronting contexts, however, the antecedent of the reflexive must be *the nanny*; in other words, in fronted *wh*-predicates, a reflexive is unambiguously cataphoric in the sense that its antecedent must occur to its right of it in the sentence. Although this has been the subject of intense research in its own right, for present purposes it suffices to note that in (3) this simply means that the reflexive inside the filler receives the same interpretation as it would as if the *wh*-predicate had never been moved (e.g., *Alice recalled that the attractive young nanny had been pleased with herself during the summer vacation*).

As mentioned above, although the ultimate interpretation of the reflexive remains the same when the predicate is fronted, this configuration could require comprehenders to adopt a different processing strategy to link the reflexive to its antecedent. Instead of relying on a backwards-looking memory retrieval, the processor might need to engage on a forward-looking search through the sentence to find an antecedent if a reflexive is encountered before its antecedent. Comprehenders engage a similar forward search process when they encounter cataphoric pronouns in both English (Coward & Cairns, 1987; Van Gompel & Liversedge, 2003; Yoshida, Kazanina, Pablos, & Sturt, 2014) and Japanese (Aoshima, Yoshida, & Phillips, 2009), and they do so in a structurally constrained way (Kazanina et al., 2007).

However, less is known about how cataphoric reflexives are processed. Frazier, Plunkett, and Clifton (1996) used a self-paced reading task to investigate the processing of *wh*-argument fronting sentences as in (4):

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- (4) **Ambiguous:** Which rumor about herself did the **matron** claim the **actress** made up?
Matrix subject: Which rumor about herself did the **actress** claim the newspaper made up?
Embedded subject: Which rumor about herself did the newspaper claim the actress made up?
-

Frazier and colleagues found that in ambiguous sentences with cataphoric reflexives, comprehenders preferred to interpret the matrix subject *matron* as the reflexive's antecedent. For the unambiguous sentences, they found that reading time was overall faster for the sentences where the reflexive matched the features of the matrix subject. Taken together, their findings suggest that in comprehension, readers preferred to associate the reflexive with the first antecedent that followed it. This finding is compatible with the hypothesis that cataphoric reflexives engage a forward-looking search through the string to find an accessible antecedent and stop their search as soon as a suitable antecedent is found (for further evidence and arguments to this effect, see Chapman, 2018).

Omaki (2010) modified the design by Frazier and colleagues to explore the extent to which the processing of dislocated reflexives is guided by the structural cues that guide the resolution of *in-situ* reflexives. His eye-tracking-while-reading experiment manipulated the type of fronted constituents as in (5) and (6), and whether the matrix subject matched or mismatched the reflexive's gender feature:

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- (5) *wh*-argument fronting
 Alice/Andrew recalled which drawing of herself the attractive young nanny had damaged __ during the summer vacation.
- (6) *wh*-predicate fronting
 Alice/Andrew recalled how pleased with herself the attractive young nanny had been __ during the summer vacation.
-

Omaki observed that processing of the reflexive was disrupted when the gender feature of the reflexive mismatched those of the antecedent in both *wh*-argument and *wh*-predicate fronting conditions. His finding suggests that in both predicate- and argument-fronting constructions, comprehenders incrementally link the reflexive to the linearly preceding matrix subject. For the potentially ambiguous *wh*-argument fronting construction (5), this observation suggests that comprehenders prefer to immediately establish the reflexive's reference, preferring the referent they have already processed (*Alice/Andrew*) to upcoming antecedents that have not yet been encountered. In other words, a backwards anaphoric search for an antecedent is preferred over a forward, cataphoric search process.

However, this behavior is unexpected in the *wh*-predicate fronting conditions, because the matrix subject *Alice/Andrew* is not a grammatically permitted antecedent for the reflexive. At the point at which the reflexive is encountered, it is yet not possible to reliably identify a grammatically licit antecedent for the reflexive, as the antecedent must follow the reflexive. For this reason, there is temporary uncertainty about which antecedent the reflexive will have: it could be an antecedent that has not yet been encountered (as in (6)), or it could be a preceding antecedent if the following material contains a pronoun that refers back to that antecedent (as in *Alice recalled how pleased with herself she had been __ during the summer vacation*). Importantly, Omaki's results show that this temporary uncertainty does not prevent comprehenders from attempting to immediately identify an antecedent for the reflexive, even though this risks creating an interpretation that violates structural constraints on reflexives. In this sense, the finding that comprehenders readily take the matrix subject as the antecedent of the reflexive seems to be at odds with the processor's relatively strict application of structural constraints on reflexives in direct object position. If structural constraints were strictly applied to guide interpretation, then we would expect comprehenders to adopt a forward search strategy to resolve the reflexive's reference, because this strategy is the only one that is guaranteed to result in a grammatical interpretation.

Omaki's finding might be taken as evidence that the pressure to assign an interpretation to the reflexive is very strong, causing comprehenders to resolve a cataphoric reflexive's reference immediately rather than engaging a forward search for its antecedent. And since that forward search is the only guaranteed strategy to form a grammatical interpretation, this finding suggests that when given the choice between immediately interpreting the reflexive and potentially violating grammatical constraints, the processor prefers the former strategy. This surprising result suggests that there is substantial pressure to resolve the reflexive's reference immediately, to the point that unambiguously cataphoric reflexives appear to be treated anaphorically in real-time processing.

The present study

The present study takes as a starting point Omaki's finding that comprehenders immediately interpret the reflexive inside a fronted *wh*-predicate, before its antecedent has been encountered. We first seek to replicate this central finding. Replicating and extending this finding would provide further evidence that comprehenders are under substantial pressure to resolve reference incrementally. In addition, we explore two hypotheses about how antecedent selection proceeds under these conditions.

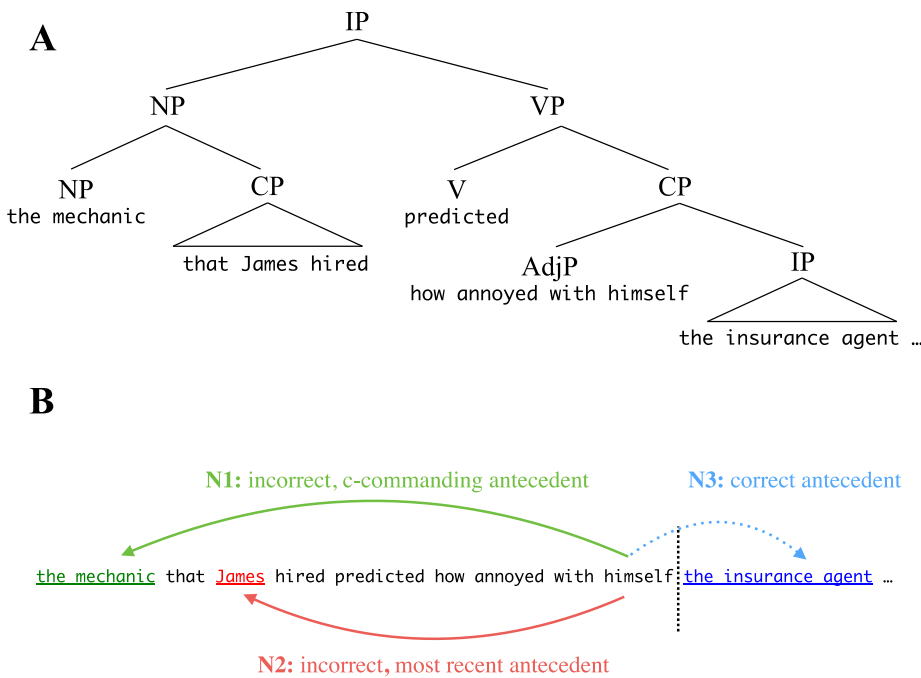


Fig. 1. Panel A: Schematic of the syntactic structure of the example sentence in (7). Panel B: The relation of the reflexive to the three intrasentential antecedents. At the point of encountering the reflexive, N1 and N2 have both been processed. The correct antecedent N3 has not yet been encountered (indicated by blue dotted line). N1 and N2 are both incorrect antecedents. N1 c-commands the reflexive. N2 does not, but it is the more recent antecedent. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

To do this, we investigate the processing of reflexives inside *wh*-predicates when there are two referents that precede the reflexive, as in (7). Fig. 1 shows the structural details of (7) with an omission of a few words for expository purposes.

(7) The **mechanic** that **James** hired this year predicted how annoyed with **himself** the extremely dumb insurance agent would be for messing up the big account.

In (7), like in the other examples considered, the ultimately correct interpretation of the reflexive is one where the reflexive refers to the embedded subject, *the extremely dumb insurance agent*. However, in a modification to the examples studied by Omaki (2010), in (7) there are two referents that have been processed before encountering the reflexive: *mechanic* and *James*.

In this scenario, there are two distinct possibilities for how comprehenders resolve the reference of the reflexive. One possibility is that the very same structural cues that restrict the antecedent search process for direct object reflexives restrict the search process in this construction as well. We call this the *structure guidance* hypothesis. On this view, the reflexive's search process in (7) will be biased towards structurally prominent (i.e. c-commanding) antecedents with the appropriate morphosyntactic features. This predicts that comprehenders will prefer to associate the reflexive with the structurally prominent subject *mechanic* over the embedded subject *James*. More broadly, this hypothesis expresses the intuition that comprehenders generally seek syntactically prominent, structurally local antecedents for reflexives, even in contexts where this is not necessary to satisfy a grammatical constraint. This would arise, for example, if the syntactically prominent subject *the mechanic* was especially salient or available in memory.

However, a second possibility is that the search process for fronted reflexives is qualitatively different from that of *in-situ* reflexives in canonical object positions. This may occur because at the point when the reflexive is encountered in (7), the processor cannot yet incrementally apply the necessary grammatical constraints to limit the search process. In this context, the retrieval of a reflexive's antecedent might rely primarily on non-structural factors. We will refer to this as the *non-structural search* hypothesis. This search mechanism could be instantiated in one of two ways. For example, the comprehender may simply choose the linearly closest antecedent because it was more recently processed and therefore potentially more active (Arnold, 2010; Cummings & Felser,

2013; a.o.). Alternatively, in the absence of structural guidance, it is possible that lexical semantic features like gender play a primary role in resolving the reflexive's reference in this context (Arnold & Griffin, 2007; Badecker & Straub, 2002; Patil et al., 2016; Parker & Phillips, 2017). If this were true, we expect comprehenders to rapidly access any antecedent that matches the gender feature of the reflexive, regardless of its structural position.

The central question that distinguishes these two hypotheses is whether comprehenders immediately link the reflexive to a structurally prominent, c-commanding subject even when there are no active grammatical constraints requiring this. We now turn to three experiments that address this question.

Experiment 1

In Experiment 1, we used eye-tracking-while-reading to investigate the time course of anaphor resolution for reflexives inside fronted *wh*-predicates. The experimental stimuli used the structure illustrated in (7). With this sentence frame, we adopted a *gender mismatch paradigm* that manipulated the fit between the gender marking on the reflexive (*himself* vs. *herself*) and the gender stereotype of N1 (*the mechanic*, male biased) or N2 (*James* vs. *Brianna*). This resulted in four conditions in a fully crossed 2 × 2 design; an example of the critical stimuli is in Table 1.

In eye-tracking-while-reading studies using the gender mismatch paradigm, evidence for reading disruption is observed when there is a mismatch between the gender of the reflexive and that of the antecedent that is initially accessed (e.g., Sturt, 2003). The structure guidance hypothesis predicts that the parser should preferentially consider the c-commanding antecedent (*the mechanic*); under this scenario, reading time on the reflexive in N1 match conditions should be reliably faster than N1 mismatch conditions.

On the other hand, the non-structural search hypothesis is compatible with two different outcomes. If it is primarily recency that determines a reflexive's reference in examples like (7), then the most recently processed referent (i.e. N2) should be considered first. In our paradigm, this will result in faster reading times for the N2 match conditions than in N2 mismatch conditions. If instead reference resolution is primarily driven by gender features, then comprehenders will rapidly access any antecedent that matches the reflexive's gender

Table 1
Sample materials and conditions for Experiment 1.

	Sentence initial	Analysis regions			Sentence final
		Pre-reflexive	Reflexive	Post-reflexive	
N1 Match: N2 Match	The mechanic that James hired this year predicted	how annoyed	with himself	the extremely dumb insurance	agent would be for messing up the big account
N1 Match: N2 Mismatch	The mechanic that Brianna hired this year predicted	how annoyed	with himself	the extremely dumb insurance	agent would be for messing up the big account
N1 Mismatch: N2 Match	The mechanic that Brianna hired this year predicted	how annoyed	with herself	the extremely dumb insurance	agent would be for messing up the big account
N1 Mismatch: N2 Mismatch	The mechanic that James hired this year predicted	how annoyed	with herself	the extremely dumb insurance	agent would be for messing up the big account
Example question	Who was thinking about the insurance agent? <i>The mechanic (or) James</i>				

feature. In this case, we expect to see a gender mismatch effect only when there is no antecedent that matches the reflexive's features; that is, we should see slowed reading times only in the *N1 Mismatch*, *N2 Mismatch* condition.

Methods

Participants

Fifty-five native speakers of American English were recruited to participate in Experiment 1. Of these, five participants were removed prior to analysis due to unacceptable levels of track loss, leaving a total of 50 participants included in the final analysis. Participants were undergraduates from the University of Massachusetts, Amherst and gave informed consent before participating. They received course credit in exchange for participation.

Materials

There were 24 critical items that crossed two independent factors of c-commanding referent (N1) match (N1 Match/Mismatch) and most recent referent (N2) match (N2 Match/Mismatch), as shown in Table 1. The target sentences consisted of a complex subject with an object gap relative clause (*the mechanic that James hired* _), a verb that takes interrogative clauses as its complements (*predicted*), a *wh*-predicate that consists of *how*, a gradable adjective (*annoyed*), and a prepositional phrase with a reflexive that specifies the object of the adjectival predicate (*with himself/herself*). The head of the complex subject (N1) consisted of an occupation name that was stereotypically biased to be either male or female according to stereotypical gender rating studies (Carreiras, Garnham, Oakhill, & Cain, 1996; Kennison & Trofe, 2003). The N2 region consisted of common male and female names that the authors judged to be strongly gender-biased. The *wh*-predicate region was followed by the embedded clause that provides the ultimate antecedent for the fronted reflexive.

The 24 target items were combined with 86 fillers. All filler sentences were of comparable length and complexity to the target items. All fillers were normatively grammatical, and contained a range of referential and structural ambiguities. Half of all items (including fillers) were followed by a simple two-choice comprehension question. Overall comprehension accuracy was 83.9% (SD = 0.07%) with a range of 67.5–98.8% out of 40 questions. Mean accuracy in the experimental items was lower than comprehension accuracy on average: mean accuracy for *N1 match*, *N2 match* was 69%, for *N1 mismatch*, *N2 mismatch* 72%, for *N1 match*, *N2 mismatch* 76%, and for *N1 mismatch*, *N2 match* 71%.

Procedure

Participants read sentences while their eye movements were recorded with a tower-mounted Eyelink 1000 eye tracker (SR Research). Participants had binocular vision while movements were measured, but only the movement of the right eye was recorded. While seated,

participants rested their chins on an adjusted chin rest and were approximately 35 in. from the monitor, giving them approximately 4 characters per 1° of visual angle. The sentences were presented one at a time on the screen in a black fixed-width font (Monaco, 11pt) on a white background. All stimuli fit on a single line of text. The experiment was conducted using the EyeTrack software developed at UMass Eyetracking Lab (<http://www.psych.umass.edu/eyelab/software>).

Before the experiment, the experimenter calibrated the eye-tracker with a 3-point display, and the experimenter ensured that eye position errors were no greater than 0.5°. Subsequent to calibration, participants were given three practice trials. Each trial was initiated after fixating on a gaze-contingency box centered along the left edge of the screen. This ensured accurate calibration before each trial; if this was difficult, recalibration was performed. Once a fixation on the box was recognized by the experimental software, the target sentence was displayed all at once. Participants used a Microsoft SideWinder controller to end a trial when they were finished reading. Then they either provided an answer to the two-alternative forced choice comprehension question, or proceeded to the next experimental trial. Questions were asked after half of all sentences. Questions probed both the content of the relative clause and the embedded predicate.

The experiment lasted no more than 45 min. After the experiment, participants were fully debriefed on the materials and given time to ask any questions. The experiment protocol for this study was approved by the Institutional Review Board at the University of Massachusetts, Amherst.

Data analysis

For purposes of analysis, the sentences were split into 5 regions as shown in Table 1. We report eye movement data from 3 regions. First, we report the pre-reflexive region (*how annoyed*) in order to ensure that there is no unexpected reading disruption immediately preceding the critical region. Second, we report the critical reflexive region (*with himself/herself*), where we expect to see reading disruptions due to gender mismatch between the reflexive and an antecedent. Third, we report the post-reflexive region (*the extremely dumb insurance*) to assess if there is a spill-over of the critical effect from the reflexive region. The reflexive region included a preposition (e.g., *with*) that immediately preceded the reflexive, which allowed us to prevent data loss due to parafoveal previews of the reflexive. The post-reflexive region included the four words following the reflexive, a size that was chosen because this excludes the head of the embedded clause subject (*agent*), so that the reading time measure in this region is not complicated by additional referential processing triggered by the head noun.

Our analyses include the following fixation measures: *first pass time*, *percent regressions out*, *go-past time*, and *total reading time* (Rayner, 1998; Rayner & Pollatsek, 2006; Staub & Rayner, 2007). The *first-pass reading time* is calculated by summing the fixations in a region between the time when the eye-gaze first enters the region from the left and the time when the eye-gaze exits the region either to the left or the right. The

percent regressions out indicate the probability that a reader made a regressive eye movement to preceding regions after fixating a given region. This measure includes only regressions made during the reader's first pass through the region, and does not include regression made after re-fixating the region.

Go-past time is the sum of fixations from the time when the eye-gaze first enters a region from the left to the time when the eye-gaze exits the region to the right. Go-past time is identical to first-pass reading time if the eye-gaze first exits the region to the right, but if the eye-gaze exits the region to the left, then go-past times are longer than the first-pass time as they include re-fixations on the current or previous regions before exiting that region to the right. *Total reading times* are calculated by summing the total durations of fixations in a region, starting with the time when the eye-gaze first enters the region and ending with the last fixation on that region.

Prior to statistical analysis, blinks and other artifacts were removed, as were any individual fixations shorter than 80 ms or 1000 ms, as were long saccades (greater than 120 ms) coming into, in, or leaving the critical reflexive region. Short fixations (< 80 ms) within one character preceding or following the critical region were combined with neighboring fixations. Trials with track loss or blinks in first-pass reading at the critical region were excluded. Participants who lost more than 25% of their data in this way were excluded from further analysis, leading to the rejection of 5 participants' data. Of the remaining participants, removal of trials with track loss led to a loss of 3.0% of the data. Lastly, additional first-pass times of greater than 2000 ms and go-past times greater than 10000 ms were excluded, as were total times of greater than 4000 ms.

While we present raw RTs in milliseconds in the text, our statistical analyses were performed on log-transformed RTs to ensure normally-distributed residuals; we used the Box-Cox transformation to verify that this transformation resulted in appropriately distributed residuals (Box & Cox, 1964). The continuous RT measures were analyzed using linear mixed effects models, using the *lme4* package in the R statistical computing environment (R Development Core Team, 2015; Bates, Maechler, Bolker, & Walker, 2015). Linear models included fixed effects terms for both *N1 match* and *N2 match*. For both fixed effects terms, we used simple difference coding, where 0.5 indicated a match between the reflexive's features and the relevant noun phrase (*N1 match/N2 match*), and -0.5 indicated a gender mismatch between the reflexive and the relevant noun phrase (*N1 mismatch/N2 mismatch*). Crossed random intercepts and uncorrelated random slopes for all fixed effect parameters were included for subject and item grouping factors (Barr, Levy, Scheepers, & Tily, 2013). To evaluate the significance of our findings, we adopt the convention that a *t* value of absolute value greater than 2 presents a significant effect at $\alpha = 0.05$ (Gelman & Hill, 2007). For analyses of the percent regressions-out measure, we used generalized logistic mixed effects models with crossed random intercepts and uncorrelated random slopes. We adopt a similar convention that a *z* value of absolute value greater than 2 presents a significant effect.

Results

Table 2 presents the raw reading times for first-pass, go-past, and total duration times, as well as percent regressions out for the three analysis regions. Table 3 summarizes the results of the mixed effects models on log-transformed reading times and the percent regressions. Fig. 2 highlights the reading measures in the critical region that are most relevant for our research questions.

In the pre-reflexive and post-reflexive regions, we observed no significant effect of our experimental manipulations on any reading measure. In the critical reflexive region, we observed a significant effect of *N2 Match* in the first pass time, go-past time and total duration. In all cases, we observed shorter reading times in the *N2 match* conditions than *N2 mismatch* conditions. There was no significant effect of *N1 Match*, nor were there any reliable interactions. Overall, this pattern

indicates that *N2 mismatch* conditions caused reliable reading disruption in multiple reading time measures. The gender fit between *N1* and reflexive did not have a reliable impact on reading times.

Discussion

Experiment 1 used a gender mismatch paradigm to evaluate how readers resolve the reference of a reflexive inside a fronted *wh*-predicate. In all three reading measures we calculated, first pass time, go-past time and total durations, we observed reading time disruption when there was a gender mismatch between the reflexive and the more recently processed *N2*. Furthermore, we did not find any evidence for an *N1 match* effect in the reading times. Thus we fail to find any evidence that comprehenders link the reflexive to the structurally prominent *N1* in this context, against the predictions of the structure guidance hypothesis. Instead this pattern provides support for the non-structural search hypothesis. The results suggest that it is recency or linear proximity, rather than match to gender features, is what guides reference resolution for reflexives in this context.

Further, this result corroborates Omaki (2010) by providing additional evidence that the parser immediately attempts to find an antecedent for a cataphoric reflexive in the preceding context, even though the grammatical antecedent must appear in the subsequent, embedded clause subject position. This is surprising given that *N2* cannot grammatically be the antecedent of the reflexive unless the embedded clause subject turns out to be a pronoun that is coreferential with *N2*. This finding suggests that the parser appears to prioritize incremental anaphor resolution over accuracy of interpretation. We will return to the potential relevance of a subsequent, coreferential pronoun in Experiment 3.

The most novel finding in this experiment is that the parser not only favors incrementality over accuracy, but also does not appear to use structural prominence or c-command to guide the antecedent search process for a reflexive in this context. This is a rather surprising observation, given the substantial evidence that the parser relies on structural information such as c-command and locality to identify antecedents for a reflexive when they occur in canonical, direct object positions. We will return to the interpretation of this finding in General Discussion.

Experiment 2

The eye movement data in Experiment 1 suggest that the antecedent search for reflexives in fronted *wh*-predicates is guided more by recency or linear proximity rather than by structural information. The goal of Experiment 2 was to partially replicate this finding, while also extending our experimental design to directly compare cataphoric and anaphoric reflexives. For anaphoric reflexives, we expect that the antecedent search process should be guided by structural principles and preferentially select the referent that c-commands the reflexive (Sturt, 2003; Xiang et al., 2009). We manipulate whether a reflexive is anaphoric or cataphoric by manipulating whether it is inside a fronted predicate phrase or is in an in-situ object position; examples of each configuration are presented in Table 4 below. By introducing a within-subjects manipulation of the reflexive's position (inside a fronted predicate vs. in-situ direct object position), we offer a more direct test of the claim that the structural position of the reflexive is the critical factor that determines the factors that guide antecedent retrieval.

For Experiment 2, the structure guidance hypothesis predicts that both in-situ and fronted reflexives should show a similar processing profile; in particular, they should both associate most readily to the prominent, matrix subject position. We tested this in experiment 2 by factorially manipulating the position of the reflexive (in-situ or fronted), and whether it matched the first or second noun in the sentence. If reflexives in both contexts access the prominent matrix position, then we should see faster reading times when *N1* matches than when *N2* matches in both in-situ and fronted configurations in Table 4.

Table 2
Experiment 1 mean reading times in milliseconds (standard error) and percent regressions.

	Pre-reflexive [“how annoyed”]	Reflexive [“with himself”]	Post-reflexive [“the extremely dumb insurance”]
<i>First-Pass Time</i>			
N1 match: N2 match	411 (15)	351 (12)	656 (29)
N1 match: N2 mismatch	394 (17)	386 (14)	649 (31)
N1 mismatch: N2 match	379 (16)	374 (14)	666 (26)
N1 mismatch: N2 mismatch	402 (18)	400 (15)	655 (29)
<i>Percent Regressions Out</i>			
N1 match: N2 match	15.2 (2.0)	10.7 (2.1)	12.9 (1.8)
N1 match: N2 mismatch	18.2 (2.4)	14.4 (2.2)	13.3 (2.5)
N1 mismatch: N2 match	13.7 (1.9)	12.7 (2.2)	11.2 (2.0)
N1 mismatch: N2 mismatch	14.8 (1.9)	14.6 (2.4)	18.8 (2.4)
<i>Go-Past Time</i>			
N1 match: N2 match	508 (20)	480 (34)	896 (62)
N1 match: N2 mismatch	520 (26)	503 (25)	868 (56)
N1 mismatch: N2 match	487 (30)	484 (30)	869 (61)
N1 mismatch: N2 mismatch	526 (34)	520 (29)	991 (48)
<i>Total Reading Time</i>			
N1 match: N2 match	727 (50)	666 (49)	1174 (84)
N1 match: N2 mismatch	696 (37)	709 (46)	1149 (71)
N1 mismatch: N2 match	666 (41)	634 (40)	1086 (65)
N1 mismatch: N2 mismatch	719 (43)	731 (48)	1194 (71)

In contrast, the non-structural search hypothesis predicts an interaction of reflexive position and N1/N2 match. In particular, reading times should be slowed when N1 mismatches for the *in-situ* reflexives and slowed when N2 mismatches for fronted *wh*-predicates.

Methods

Participants

We recruited 44 native speakers of American English from the Johns Hopkins University community, but the data from one participant were removed due to excessive track loss. Additionally, participants with lower than 65% accuracy on comprehension questions were removed from analyses, leaving a total of 39 participants in the final analysis. Participants gave informed consent and received course credit or \$10 for their participation. Completion of the experiment, including calibration of the eye-tracker, took approximately 45 min.

Materials

We used 24 sets of target items, which factorially manipulated the reflexive position (*in-situ* vs. *fronted*) and antecedent match (N1 match/N2 mismatch vs. N1 mismatch/N2 match). Table 4 presents a sample set.

The fronted predicate conditions were constructed from items in Experiment 1, with a few modifications. First, we took only the conditions in which either N1 or N2 matched the gender of the reflexive. Second, we added a Prepositional Phrase modifier after the reflexive, which served as a lexically matched spill-over region across conditions. The *in-situ* predicate counterparts were created from the fronted predicate conditions. More specifically, we deleted the main clause verb (e.g., *realized*) in the sentence initial region, changed the *wh*-phrase (e.g., *how mad*) to a copula plus adjective (e.g., *was mad*) in the pre-reflexive region, and substituted a modifier phrase for an embedded clause in the sentence final region. Due to the extra verb and an additional level of clausal embedding in the fronted predicate conditions, we expected that reading time measures may be generally inflated in

Table 3
Summary of model estimates, standard errors (SE), and *t*-values (first pass time, go-past time and total duration) and *z*-values (percent regressions out) in Experiment 1. Bolded values indicate significant effects at $\alpha = 0.05$.

	Pre-reflexive		Reflexive		Post-reflexive	
	Estimate (SE)	<i>t</i> or <i>z</i>	Estimate (SE)	<i>t</i> or <i>z</i>	Estimate (SE)	<i>t</i> or <i>z</i>
<i>First-Pass Time</i>						
N1	0.03 (0.03)	1.10	−0.04 (0.03)	−1.43	−0.02 (0.03)	−0.54
N2	0.01 (0.03)	0.24	−0.06 (0.03)	−2.21	0.02 (0.03)	0.76
N1*N2	0.1 (0.06)	1.54	−0.01 (0.05)	−0.13	−0.04 (0.06)	−0.75
<i>Percent Regressions Out</i>						
N1	0.15 (0.17)	0.91	−0.08 (0.19)	−0.42	−0.12 (0.17)	−0.70
N2	−0.15 (0.17)	−0.91	−0.27 (0.22)	−1.25	−0.35 (0.18)	−1.96
N1*N2	−0.15 (0.33)	−0.44	−0.19 (0.42)	−0.46	0.55 (0.38)	1.44
<i>Go-Past Time</i>						
N1	0.03 (0.03)	1.04	−0.03 (0.03)	−0.87	−0.04 (0.04)	−1.19
N2	−0.01 (0.03)	−0.32	−0.08 (0.04)	−2.26	−0.05 (0.03)	−1.53
N1*N2	0.05 (0.06)	0.80	−0.04 (0.07)	−0.60	0.11 (0.07)	1.73
<i>Total Reading Time</i>						
N1	0.02 (0.03)	0.72	0 (0.03)	0.12	0 (0.03)	0.02
N2	−0.03 (0.03)	−0.99	−0.1 (0.04)	−2.52	−0.05 (0.03)	−1.66
N1*N2	−0.09 (0.06)	1.50	−0.05 (0.06)	0.82	0.11 (0.07)	1.51

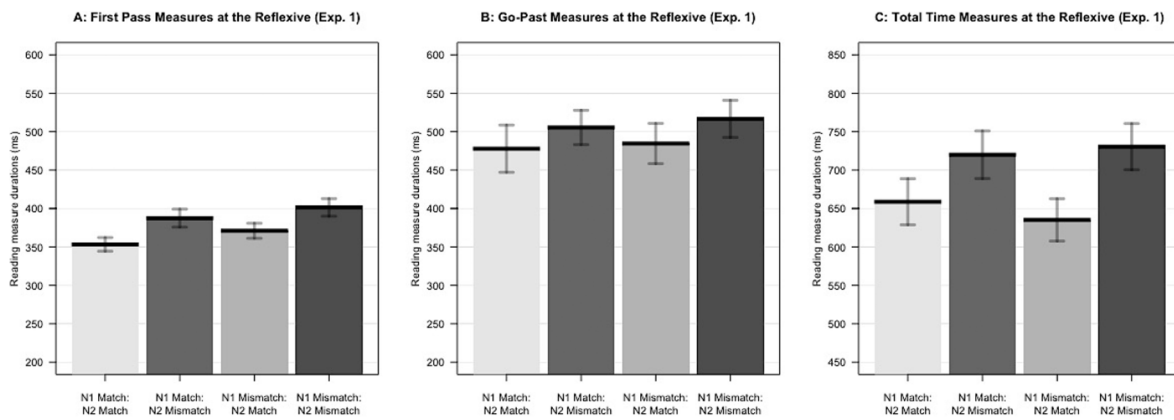


Fig. 2. Average First pass (A), Go-Past (B) and Total reading time (C) measures with standard errors at the critical reflexive region in Experiment 1.

the fronted predicate conditions.

The 24 sentence sets were counterbalanced across four lists so that each participant saw only one version of the target items and consequently read six tokens of each condition. In addition, 60 fillers of similar length and complexity were constructed and added to each list. All fillers were grammatical, and contained a range of referential and structural ambiguities. All items (including fillers) were followed by a simple two-choice comprehension question. Overall comprehension accuracy was 83.2% (SD = 0.07%) with a range of 67.1–100.0%. Mean accuracy for the in-situ conditions was 91% for *N1 match*, *N2 mismatch* and 84% for the *N1 mismatch*, *N2 match*. For the fronted conditions, it was 72% for *N1 match*, *N2 mismatch* and 84% for the *N1 mismatch*, *N2 match*.

Procedure

An Eyelink 1000 eye-tracker (SR Research: Mississauga, ON, Canada) was used to record eye movements. The participant's head was stabilized by a chin rest and a forehead rest. Participants had binocular vision while eye movements were measured, but only movement of the right eye was recorded at a sampling rate of 1000 Hz. The stimulus presentation was controlled via Experiment Builder software from SR Research. The eye-tracker display allowed a maximum of 137 characters per line, in 9pt fixed-width Monaco font. Some filler sentences were displayed on two lines, but all target sentences were displayed on one line. Stimuli were displayed on a 21.5-in. Samsung SyncMaster monitor, and participants were seated 65 cm from the computer screen, giving them approximately 4 characters per 1° of visual angle. Before the experiment started, participants were seated in front of the eye-tracker and received instructions for the experiment. A 9-point calibration routine was performed at the beginning of the experiment, and the experimenter monitored the calibration accuracy throughout the session, recalibrating when necessary to ensure that eye position errors were no greater than 0.5°. The experiment then began with written

instruction on the display and five practice trials.

At the beginning of each trial, a black gaze-contingency circle was displayed on the left side of the monitor, which corresponded to the location of the beginning of the sentence. The text was displayed only after the participant successfully fixated on the circle. After reading each sentence, the participant pressed a button on a controller to remove the sentence display. Each sentence was followed by a two-alternative forced choice comprehension question, and the participant answered the comprehension question by pressing a left or right button. Comprehension questions probed both the content of the relative clause and the embedded predicate, but never addressed the critical reflexive region in the sentence. The entire experiment lasted no more than 45 min. After the experiment, participants were fully debriefed on the materials and given time to ask any questions. Data were then post-processed using Eyelink Dataviewer software from SR Research. The experiment protocol for this study was approved by the Homewood Institutional Review Board at the Johns Hopkins University.

Data analysis

We used the same data analysis and exclusion procedures as Experiment 1. The same set of fixation measures (first pass time, go-past time, percent regressions out and total reading time) in the 3 critical regions (pre-reflexive, reflexive, post-reflexive) were computed (see Table 4 for a representation of the regions selected). Participants who lost more than 25% of their data due to track loss were excluded from further analysis, leading to the rejection of one participants' data. Additionally, first-pass times greater than 2000 ms and go-past times greater than 10000 ms were excluded, as were total times of greater than 4000 ms. In the statistical analyses of the fixation measures, we again used simple difference coding for our fixed effects. In the present design, these were antecedent match (*N1 Match* = 0.5; *N2 Match* = −0.5), and reflexive position (fronted predicate = 0.5; in-situ predicate = −0.5), as well as their interaction.

Table 4
Sample materials and conditions for Experiment 2.

		Analysis regions			
	Sentence initial	Pre – reflexive	Reflexive	Post-reflexive	Sentence final
In-situ predicate, N1 match, N2 mismatch	The nanny that Steven hired last week	was mad	at herself	for making a mess	after playing with the child
In-situ predicate, N1 mismatch, N2 match	The nanny that Steven hired last week	was mad	at himself	for making a mess	after playing with the child
Fronted predicate, N1 match, N2 mismatch	The nanny that Steven hired last week realized	how mad	at herself	for making a mess	the sloppy child would be
Fronted predicate, N1 mismatch, N2 match	The nanny that Steven hired last week realized	how mad	at himself	for making a mess	the sloppy child would be
Example question	Who was hired? <i>The nanny (or) Steven</i>				

Table 5

Experiment 2 participant mean reading times in milliseconds (standard error) and percent regressions.

	Pre-reflexive [“was mad/how mad”]	Reflexive [“at himself”]	Post-reflexive [“for making a mess”]
<i>First-Pass Time</i>			
in-situ, N1 match, N2 mismatch	336 (16)	311 (13)	616 (25)
in-situ, N2 match, N1 mismatch	336 (13)	333 (15)	640 (26)
fronted, N1 match, N2 mismatch	337 (14)	340 (16)	629 (24)
fronted, N2 match, N1 mismatch	346 (14)	325 (15)	626 (29)
<i>Percent Regressions Out</i>			
in-situ, N1 match, N2 mismatch	8.9 (2.1)	12.7 (2.9)	11.3 (2.2)
in-situ, N2 match, N1 mismatch	9.3 (2.4)	12.3 (2.7)	13.2 (2.2)
fronted, N1 match, N2 mismatch	16.4 (2.8)	16.1 (3.1)	20.7 (2.7)
fronted, N2 match, N1 mismatch	11.8 (2.7)	13.4 (2.9)	20.9 (3.5)
<i>Go-Past Time</i>			
in-situ, N1 match, N2 mismatch	393 (18)	408 (31)	745 (32)
in-situ, N2 match, N1 mismatch	396 (19)	436 (34)	830 (42)
fronted, N1 match, N2 mismatch	447 (25)	461 (32)	956 (85)
fronted, N2 match, N1 mismatch	416 (22)	409 (27)	968 (79)
<i>Total Reading Time</i>			
in-situ, N1 match, N2 mismatch	486 (27)	467 (28)	884 (45)
in-situ, N2 match, N1 mismatch	571 (35)	565 (35)	1033 (45)
fronted, N1 match, N2 mismatch	681 (42)	649 (43)	1248 (72)
fronted, N2 match, N1 mismatch	657 (44)	642 (49)	1230 (67)

Results

Table 5 presents the participant means on each fixation measure for each region as well as the standard errors of the participant means. Table 6 presents a summary of the statistical analyses. Fig. 3 highlights the reading measures in the critical region that are most relevant for our research questions.

In the pre-reflexive region, the percent regression and total reading time measures showed a main effect of reflexive position, with longer reading times and greater percent regression in the fronted predicate conditions than in the in-situ predicate conditions. This effect was expected because the sentence initial region in the fronted predicate conditions contained the extra verb, which made it more likely to attract multiple fixations and regressions in the immediately following pre-reflexive region.

In the critical reflexive region, an interaction between antecedent match and reflexive position was found in the go-past time measure. This interaction pattern indicated that in the in-situ predicate conditions, reading times were shorter when the reflexive matched in gender with N1 (408 ms for N1 match / N2 mismatch, and 436 ms for N2

match / N1 mismatch). However, in the fronted predicate conditions, the pattern was numerically reversed: reading times were shorter when the reflexive matched in gender with N2 (461 ms for N1 match and 409 ms for N2 match in wh conditions). To resolve this interaction, we fit a second linear mixed effects model to the go-past times at the critical region, with separate contrasts for antecedent match within the fronted conditions and in-situ conditions. This model revealed a significant effect of antecedent match for fronted conditions ($\beta = -0.05$ (± 0.02), $t = -2.1$), but no reliable effect for in-situ conditions ($\beta = 0.02$ (± 0.03), $t = 0.6$).

The total reading time measure in the reflexive region revealed a main effect of reflexive position. This is likely to reflect the same effect observed in the pre-reflexive region, which is due to the presence of an embedding verb in the sentence initial region. We also found an interaction of antecedent match and reflexive position, which was driven by the presence of a GMME in in-situ predicate conditions (467 ms for N1 match / N2 mismatch, and 565 ms for N2 match / N1 mismatch), and its absence in the fronted predicate conditions (649 ms for N1 match / N2 mismatch, and 642 ms for N2 match / N1 mismatch). As

Table 6Summary of model estimates, standard errors, and *t*-values (first pass time, go-past time and total reading time) and *z*-values (percent regressions out) for four eye movement measures in Experiment 2 (logRTs). Bolded values indicate significant effects at $\alpha = 0.05$.

	Pre-reflexive		Reflexive		Post-reflexive	
	Estimate (SE)	<i>t</i> or <i>z</i>	Estimate (SE)	<i>t</i> or <i>z</i>	Estimate (SE)	<i>t</i> or <i>z</i>
<i>First-Pass Time</i>						
Match	−0.04 (0.03)	−1.24	−0.01 (0.03)	−0.17	−0.02 (0.03)	−0.57
Position	0.01 (0.03)	0.32	0.02 (0.03)	1.00	0 (0.03)	−0.02
Match*Position	−0.06 (0.05)	−1.01	0.11 (0.06)	1.70	0.04 (0.06)	0.66
<i>Percent Regressions Out</i>						
Match	0.19 (0.24)	0.81	0.12 (0.21)	0.56	−0.09 (0.26)	−0.35
Position	0.55 (0.23)	2.42	0.21 (0.21)	1.00	0.71 (0.28)	2.56
Match*Position	0.49 (0.47)	1.03	0.24 (0.46)	0.53	0.22 (0.41)	0.52
<i>Go-Past Time</i>						
Match	0 (0.03)	0.07	0.03 (0.03)	0.83	−0.05 (0.05)	−0.98
Position	0.06 (0.04)	1.69	0.06 (0.03)	1.78	0.12 (0.05)	2.19
Match*Position	0.04 (0.06)	0.69	0.15 (0.07)	2.14	0.07 (0.07)	1.04
<i>Total Reading Time</i>						
Match	−0.05 (0.04)	−1.49	−0.06 (0.04)	−1.48	−0.08 (0.04)	−2.13
Position	0.21 (0.04)	6.09	0.21 (0.04)	5.17	0.24 (0.04)	5.81
Match* Position	0.13 (0.08)	1.73	0.2 (0.08)	2.56	0.15 (0.06)	2.43

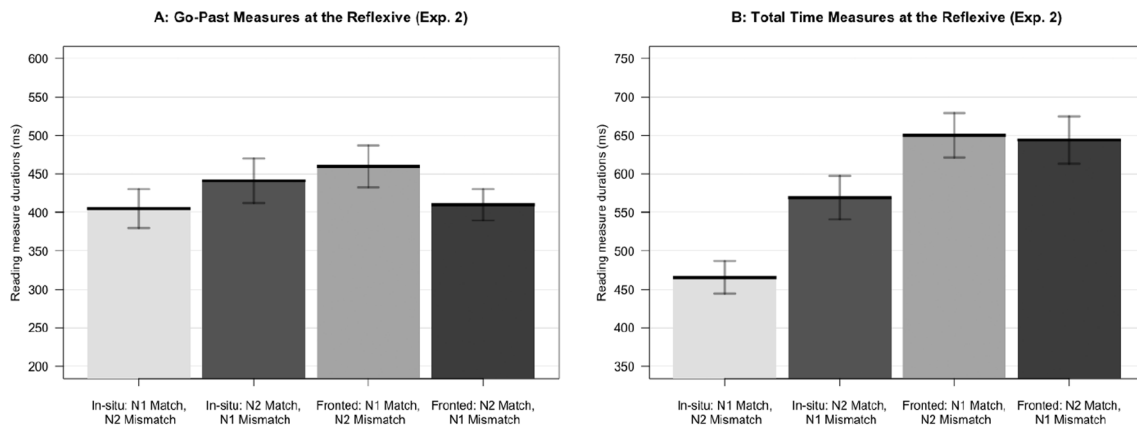


Fig. 3. Average Go-Past (A) and Total Reading Time (B) measures with standard errors at the critical reflexive region in Experiment 2.

before, we fit a second linear mixed effects model to resolve this interaction. There was no reliable effect of antecedent match for fronted conditions ($\beta = -0.02$ (± 0.03), $t = -0.6$), but we did see a reliable effect of antecedent match for in-situ conditions ($\beta = 0.08$ (± 0.03), $t = 2.7$).

In the post-reflexive region, percent regressions out, go-past time and total durations showed a main effect of reflexive position with longer reading times or more regressions for fronted predicate conditions. In total duration measures, there was again a significant interaction between reflexive position and antecedent match. The interaction pattern here resembles the one found in the total duration in the reflexive region, reflecting the presence of gender mismatch effect in in-situ predicate conditions and its absence in fronted predicate conditions. Resolving this interaction as before revealed no reliable effect of antecedent match for fronted conditions ($\beta = 0.00$ (± 0.02), $t = 0.1$), but a reliable effect for in-situ conditions comparable in magnitude to the effect seen at the critical region ($\beta = 0.08$ (± 0.03), $t = 2.7$).

Discussion

The main findings in Experiment 2 can be summarized as follows. First, we found generally longer reading times and increased regressions in the fronted predicate conditions; this is plausibly due to the difference in the level of clausal embedding or local structural complexity in these conditions. Importantly, however, we found a significant interaction between the reflexive's syntactic position and antecedent match in the go-past time; this interaction indicated the presence of two, opposing GMME depending on the reflexive position. Specifically, when the reflexive mismatched N1 gender in the in-situ predicate conditions, reading time was slowed. In contrast, when the reflexive mismatched N2 gender in the fronted predicate conditions, reading time was slowed. The finding of a gender mismatch effect when the reflexive mismatched the more recent N2 broadly replicates the finding in Experiment 1, as well as Omaki (2010). However, it is important to note that unlike in Experiment 1, the full crossover interaction was only observed in a single measure, go-past time. In this sense, we did not replicate the exact pattern of significance seen in Experiment 1, and correspondingly the evidence for this effect in Experiment 2 is weaker. In isolation, a significant effect in a single eye-tracking measure when multiple measures and regions are tested may not provide strong evidence for an effect, owing to a multiple comparisons problem (von der Malsburg & Angele, 2017). There is clearer evidence, however, for the interaction of reflexive position and antecedent match, as this effect was observed in three different measures. This critical interaction pattern is consistent with the hypothesis that antecedent search is biased towards c-commanding referents when reflexives appear in canonical object positions, but that this is relaxed when reflexives appear in non-canonical, fronted positions, such that recently processed referents become more prominent and more accessible.

In addition, we observed an interaction of reflexive position and antecedent match in total reading times at both the critical and spillover regions. The shape of this interaction was somewhat different from that seen at go-past times at the critical region. Specifically, this interaction appeared to be driven by a strong GMME for N1 mismatch in the in-situ conditions, with little to no N1 match effect for the fronted conditions. While this finding is broadly consistent with our claim that the accessibility of the c-commanding referent is diminished in the fronted *wh*-predicate conditions, it does suggest other processing differences between in-situ and fronted reflexives. One possible explanation is that readers rapidly recognize that the link between a fronted reflexive and the most recent noun is not grammatically licensed, allowing them to reject this interpretation of the reflexive. If participants can quickly realize that they must instead engage a cataphoric search for the reflexive's antecedent (Chapman, 2018; Frazier et al., 1996), it may minimize the gender mismatch penalty observed in the reading times for *wh*-predicate conditions. While this is a plausible hypothesis, this is at present a *post-hoc* speculation, and we leave this question open for future research.

While these findings lend support to the hypothesis that the structural position of reflexives modulates the antecedent search mechanism, there is at least one alternative explanation for the N2 mismatch effect we saw for reflexives in fronted *wh*-predicates. Our explanation has so far characterized the critical process as an antecedent retrieval mechanism, but it is possible that the same result is obtained through an antecedent *prediction* process. Recall that a reflexive inside a fronted *wh*-predicate is grammatically constrained to refer to the embedded subject. Suppose that as soon as readers encounter fronted *wh*-predicates (e.g., *how mad...*), they predict the lexical content of the upcoming, embedded clause subject, prior to encountering the reflexive. This prediction would yield an accessible antecedent for the reflexive. If this subject prediction is somehow biased towards introducing a pronoun (e.g., *he*) that refers to recently processed N2 (*Steven*), then encountering *herself* would cause reading disruption, as observed in Experiments 1 and 2. On this alternative account, the N2 mismatch effect does not reflect antecedent search mechanisms that are structurally unconstrained; instead, the effect reflects lexical and syntactic expectations that sentence comprehenders generate in our sentences.

Experiment 3

Experiment 3 was designed to address the alternative explanation that pronoun prediction may be driving the eye movement results from the previous experiments. The key claim of this alternative account is that participants are making predictions about the upcoming embedded clause subject, and more specifically, that this prediction is that the embedded subject corefers with the subject of the relative clause (N2).

Table 7
A sample item set for Experiment 3.

Conditions	Sentence example
No Reflexive	The mechanic that Brianna hired this year wondered how annoyed with ...
N1 Match	The mechanic that Brianna hired this year wondered how annoyed with himself ...
N2 Match	The mechanic that Brianna hired this year wondered how annoyed with herself ...

If these two conditions are met, then we might explain the N2 match effect in eye-tracking as a side effect of this lexical expectation.

To evaluate the plausibility of this alternative explanation, we conducted a sentence completion task in order to determine comprehenders' expectations about how the sentence would complete after the critical, fronted *wh*-predicate. We were particularly interested in estimating the probability of producing pronouns that coreferred with either N1 or N2. To this end, three sentence preambles were constructed using the N1 match and N2 match fronted-predicate conditions from Experiment 1 (see Table 7). Each of the three preambles was constructed from these two conditions in Experiment 1, yielding three conditions: (1) a preamble that goes up to the preposition and does not include a reflexive (*No Reflexive* condition), (2) a preamble that includes a reflexive that refers to the matrix subject (*N1 Match* condition), and (3) one that has a reflexive that refers to the embedded relative clause subject (*N2 Match* condition). In all three conditions, the preambles were identical up through the preposition that preceded the reflexive, and N1/N2 had opposite gender biases.

This completion task allows us to assess the predictions of the antecedent prediction account. If the sentence context prior to the reflexive is sufficient for predicting a pronoun that matches N2, we expect to see a high rate of continuations that include a N2 matching pronoun (e.g., *she* for *Brianna*) in the *No Reflexive* condition. However, if this type of pronoun prediction is possible only after encountering the reflexive, then we expect to see a high rate of N1 matching pronoun production in N1 Match reflexive condition, and a high rate of N2 matching pronoun production in N2 Match reflexive condition.

The completions in the *No Reflexive* condition are critical for assessing the plausibility of the pronoun prediction hypothesis of our eye-tracking data. We reasoned that predictive processes would impact the reading of the reflexive only if those predictions were made prior to encountering the reflexive. Therefore, it is of critical importance to know what predictions are generated independent of the reflexive pronoun. If comprehenders anticipate pronouns only *after* encountering the reflexive, then it is less clear that predictive processes can satisfactorily explain our eye-tracking data, because in this situation the predicted structure would not be in place early enough to guide the processing of the reflexive itself. In other words, predictions that are conditioned on the presence of the reflexive could not guide the early processing of the reflexive as measured in our online measures from Experiments 1 and 2. Thus, an antecedent prediction explanation for the eye movement data in Experiments 1 and 2 critically requires pronoun prediction to occur *prior* to encountering the reflexive.

Methods

Participants

Forty-four native speakers of English were recruited using Amazon's Mechanical Turk (<https://www.mturk.com>). The task took approximately 20 min. Participation was restricted to workers with IP addresses in the United States, and all 44 participants self-reported as native speakers of English in a demographic survey. Participants gave informed consent and were paid \$2 for their participation in the experiment.

Materials

Sentence completion tasks have been used to investigate lexical and

syntactic prediction (e.g., Aoshima, Phillips, & Weinberg, 2004; Federmeier & Kutas, 1999; Kazanina et al., 2007; Staub, Grant, Astheimer, & Cohen, 2015) and have been shown to be effective on Mechanical Turk (Schnoebelen & Kuperman, 2010). Participants were presented with incomplete sentence preambles that were truncated forms of the *wh*-predicate stimuli used in Experiment 1. From these original 24 experimental items, we created items with three types of preamble (see Table 7). The preamble either had no reflexive pronoun (*No Reflexive* condition), or had a reflexive pronoun that referred to the matrix subject (N1 Match condition), or the relative clause subject (N2 Match condition). All three preambles had *wh*-fronted predicates that ended in '*with*' or the associated reflexive pronoun (*himself*, *herself*). The experimental items were combined with 40 filler sentences. All fillers were in the past tense and were made to be similar to the target sentences in terms of complexity and length. Only 24 of the filler sentences contained *wh*-predicates. The 24 critical experimental items were distributed into three Latin Square lists and combined with the 40 filler sentences described above. The order of presentation was randomized individually for each participant.

Procedure

This task was administered over the Internet using the IboxFarm experimental software (<http://spellout.net/ibexfarm>), which randomized the list number and item presentation. Each participant saw 8 observations per condition. Participants were instructed to finish incomplete sentences by typing a continuation in the provided box. Participants were given additional instructions to respond naturally and instinctively. There were four practice items to help introduce them to the task. The experimental and filler items were presented one after another with a 500 ms wait screen in between. The task took no more than 20 min to complete.

Data analysis

Two independent coders analyzed the completion data to the target preambles to determine how participants finished the preambles. First, these responses were coded with respect to grammaticality. Both coders showed a high degree of agreement with respect to the grammaticality of continuations: they agreed in 99.5% of cases (Cohen's $\kappa = 0.97$). If either coder decided a completion was ungrammatical, it was removed from the statistical analyses reported below. In addition, two participants were excluded before statistical analyses for having more than 25% ungrammatical completions. The mean percentage of ungrammatical completions per participant was 4.27% with a range of 0–25% ungrammatical (SD = 5.71%).

Responses were classified into four categories: (1) pronoun with N1 antecedent (the matrix subject), (2) pronoun with N2 antecedent (the RC subject), (3) referring expressions that introduce a new discourse referent that is not N1 or N2 or (4) "other" responses, which were ungrammatical continuations. This coding was done in two ways. On the *lenient* coding criterion, all continuations were treated as if they were unambiguous. Each coder independently made a categorical determination of the pronoun's referent, following a criterion that in order to refer to N1, a pronoun must match N1 in stereotypical gender. For example, this means that only completions with *she* were treated as potentially coreferent with a stereotypically female N1 like *the fashion model*. The only exception to this rigid criterion was when the continuation clearly disambiguated the referent (e.g., *The carpenter that*

Table 8
Summary of coding scheme for the completion task.

Response Type	Sentence preamble	Provided continuation
(1) Pronoun with N1 antecedent	The <u>plumber</u> that Kayla hired on Monday considered how annoyed with <u>himself</u>	he had been for not finding the ring Kayla had lost in the drain
(2) Pronoun with N2 antecedent	The mechanic that <u>Brianna</u> hired this year wondered how annoyed with	her new car she was.
(3) Referring expressions	The <u>beautician</u> that <u>William</u> consulted while shopping recollected how unhappy with <u>himself</u>	<u>the model</u> was after shaving his head
(4) Ungrammatical	a) The <u>cheerleader</u> that <u>Richard</u> befriended this summer knew how pleased with <u>herself</u> b) The <u>cheerleader</u> that <u>Richard</u> befriended this summer knew how pleased with	for making the team he was because she spent so much time with him

Haley consulted about the deck anticipated how ashamed of herself...she'd be for forgetting to send Haley a follow-up call). In these cases, the coders followed the continuation, and treated the pronoun as referring to N1 even though it does not match the intended stereotypical gender. Following this lenient coding scheme, the independent coders were in agreement about the antecedent's referent 99.4% of the time (Cohen's $\kappa = 0.99$) (see Table 8).

In addition to the *lenient* coding criterion, we also employed a *strict* coding criterion in which we removed all potentially ambiguous completions prior to analysis. On the strict coding criterion, each coder independently made a judgment as to whether the pronoun was potentially ambiguous. For example, although *he* does not match our normed stereotypical gender for an N1 like *the fashion model*, there were several examples in the data that suggested participants referred to N1 using the non-dominant stereotypical gender. One such an example is *The fashion model that Mark partied with last week imagined how ashamed of himself...he had been for slipping on the runway*. Unlike the unambiguous *carpenter* example, the provided response does not fully disambiguate the referent. There was a high rate of agreement about potential ambiguity; in 88.6% of examples, raters agreed about whether the pronoun was potentially ambiguous (Cohen's $\kappa = 0.76$). In cases of disagreement, a completion was labeled ambiguous if either coder rated it as ambiguous in order to be maximally conservative. According to this criterion, 26% of completions were considered ambiguous and removed from further analysis. Thus, for the *lenient* coding criterion, we had 1008 completions in the analysis reported below (336 in each condition), and for the *strict* coding criterion, we had 750 (280 in the No Reflexive condition, 331 in the N1 Match condition, and 139 in the N2 Match condition).

As the critical measure for testing the antecedent prediction account comes from the rate of N2 match pronoun prediction, the completion data were transformed into discrete binary measures: 1 for all completions with N2 antecedents, and 0 for all N1 completions. In order to directly test N1 and N2 completion rates across the preamble types (*No Reflexive*, *N1 Reflexive*, and *N2 Reflexive*), we excluded all irrelevant response types (*Referring expressions* and *Ungrammatical*) from the analyses. For analyses of these discrete measures, we used a generalized logistic mixed effects model with crossed random intercepts only, as the maximal model did not converge. For the analysis, dummy coding was used to contrast the baseline *No Reflexive* condition to both *N1 Match* and *N2 Match Reflexive* conditions. In this model, the intercept coefficient represented the log odds of an N2 completion over an N1 completion for the critical *No Reflexive* condition. Thus, an intercept that is

significantly different from 0 in this model would indicate a preference for either N1- or N2-referring completions (depending on sign) in the *No Reflexive* condition. Significant effects are indicated by a z value of absolute value greater than 2 (Gelman & Hill, 2007). Models were fit separately to the data from the lenient and strict coding.

Results

The completion data proportions can be seen in Table 9. In the *N1 Match Reflexive* condition, there was a clear preference for pronoun continuations with N1 antecedents (89% on the lenient coding criterion, 95% on the strict criterion), and similarly, in the *N2 Match Reflexive* conditions, there was a clear preference for continuations with a pronoun that matched N2 (85% on lenient coding, 78% on strict coding). In the *No Reflexive* preamble condition, we saw no clear preference for N1 (38%) over N2 (36%) pronoun continuations in the lenient coding criterion. For the strict coding criterion, we saw a slight N1 preference emerge in the *No Reflexive* preamble condition (46% N1 completions to 27% N2 completions).

In the *No Reflexive* condition, we also note that participants provided a reflexive in the fronted *wh*-predicate approximately 23% of the time (80 cases). If a participant provided a reflexive pronoun in the *wh*-predicate, it matched N1 in a majority of cases (55%; 44 observations), followed by N2 (36%; 29 observations). The remainder of the reflexives examples did not refer to either N1 or N2.

In order to examine in detail the distribution of N1 and N2 matching pronouns, we isolated only those completions that provided a pronoun with an N1 or N2 match. Table 10 reports the results from the generalized logistic mixed models for both the lenient and the strict coding criteria. There are significant main effects of having either the *N1 Match* or *N2 Match Reflexives* on the proportion of N2 match pronoun production, for both models. This corroborates the pattern seen in the data above, and confirms that there is a strong offline preference to link the reflexive pronoun inside the fronted-*wh* predicate to the embedded subject position. There is a significant decrease in the amount of N2 match pronouns produced when the preamble includes an N1 match reflexive. This is attributed to the increase of N1 match pronouns used when the preamble provided a reflexive matching the N1 gender feature. In the opposite direction, there is a significant increase in the use of N2 match pronoun completions when the provided reflexive matches the gender feature of the N2. In sum, these results demonstrate that selective pronoun prediction for either N1 or N2 antecedents is only possible after encountering reflexives that provide gender information.

Of critical interest was the *No Reflexive* condition. On the lenient

Table 9
Proportion of completion type for each preamble type in Experiment 3 under the lenient coding criterion. Strict coding criterion given in parentheses.

Preamble Type	Pronoun with N1 antecedent	Pronoun with N2 antecedent	Referring expressions	Ungrammatical
(A) <i>No Reflexive</i>	0.38 (0.46)	0.36 (0.27)	0.11 (0.13)	0.15 (0.14)
(B) <i>N1 Match Reflexive</i>	0.89 (0.95)	0.02 (0.00)	0.03 (0.03)	0.06 (0.02)
(C) <i>N2 Match Reflexive</i>	0.05 (0.09)	0.85 (0.78)	0.04 (0.11)	0.05 (0.01)

Table 10

Summary of generalized linear mixed effects model estimates, standard errors, and z-values for the discrete coding on the completion task data (1 for referring to N2, 0 for N1 completions). Each preamble type was contrasted to the baseline with no reflexive pronoun provided.

Preamble Type	Lenient coding estimate (SE)	z	Strict coding estimate (SE)	z
No Reflexive (Intercept)	−0.05 (0.17)	−0.30	−0.65 (0.23)	−2.81
N1 Match Reflexive	−4.06 (0.44)	−9.17	−5.39 (1.01)	−5.29
N2 Match Reflexive	2.96 (0.29)	10.29	2.79 (0.36)	7.78

coding criterion, we failed to observe any reliable difference between N1 and N2 completions (indicated by the intercept coefficient, which was not reliably different than 0). There was no evidence, then, for an N2 completion bias under the lenient coding. On the strict coding, we did observe an intercept that was reliably different than zero, but negative: this indicates a significant N1 completion bias when only the unambiguous continuations were considered.

Discussion

In Experiment 3, we saw high rates of pronoun production in all conditions. This suggests that in general, comprehenders may have expected that the embedded clause would continue with a pronoun that is coreferent with an already-mentioned entity. In conditions that presented a gender-biased reflexive in the preamble, we saw very high rates of continuations with a coreferent pronoun with the reflexive. However, in the critical No Reflexive condition, the proportions of N1 match pronouns and N2 match pronouns were almost identical on the lenient coding criterion, and biased in favor of N1 in the strict coding criterion. Thus, to the extent that there is any evidence in favor of a bias completion in our No Reflexive condition, it is in favor of N1, not N2. In this sense, these data show that there is no bias towards predicting an N2 match pronoun in our materials; if anything, they show the opposite preference.

We believe that these data do not provide clear support for a prediction-oriented explanation of the findings in our eye-tracking experiments. Assuming a linking hypothesis whereby the slowdown seen at the reflexive is a function of how well its features match the predicted antecedent, then it must be the case that there is a stronger prediction for a pronoun linked to N2 to correctly account for our finding that reading time is slower when N2 mismatches the reflexive's features than when N1 mismatches. On this view, this slowdown would indicate a more substantial mismatch between the strongly predicted features of N2 and what is encountered on the reflexive. However, if readers predict N1 and N2 linked pronouns in equal proportion, then reading times at the reflexive should be identical in both the N1 and N2 mismatch conditions, because the reflexive mismatches the predicted pronoun's features in both cases to an equal degree. Our cloze data indeed suggest that comprehenders predict N1 and N2 linked pronouns in equal proportion (or, if anything, our data suggest a bias in favor of sharper predictions for N1). This fails to provide any support the prediction-oriented explanation of our online data, as this would have been most cleanly supported by a clear bias in favor of continuations with pronouns linked to N2.

Overall, the results of Experiment 3 suggest that the predictions about the contents of the embedded clause do not clearly provide an alternative explanation for the N2 (or recency) bias that we saw in our eye-tracking data. In this sense, these findings are consistent with our claim that the behavior seen in our online eye-tracking measures reflects an antecedent retrieval process triggered by the reflexive that results in the retrieval of the linearly closer N2 antecedent.

General discussion

In two eye-tracking while reading experiments, we found that cataphoric reflexives are processed differently than anaphoric reflexives. We replicated previous findings that direct object anaphoric reflexives

showed an immediate GMME when they mismatched the features of the local subject of their clause (Experiment 2). In contrast, cataphoric reflexives inside fronted *wh*-predicates instead showed a GMME effect when they mismatched features of the linearly closer, less syntactically prominent N2 position (Experiment 1 and Experiment 2). Overall, these findings confirm Omaki (2010) core observation: readers in both Experiment 1 and Experiment 2 attempted to anaphorically interpret cataphoric reflexives in fronted *wh*-predicates, searching for an antecedent for the reflexive in the preceding context rather than engaging a forward search for an antecedent. This pattern suggests that comprehenders prefer to immediately resolve a cataphoric reflexive's reference by retrieving an antecedent in the previous context, even though this introduces the risk of a potentially incorrect interpretation. This suggests that there may be substantial pressure to resolve the reflexive's reference immediately.

Even though readers seemed to incrementally interpret reflexives in fronted *wh*-predicate constructions, we also observed evidence that they process these reflexives differently from anaphoric reflexives in canonical object position. In our experiments, we found that readers prefer to associate reflexives inside of fronted *wh*-predicates with the most recently processed antecedent (Experiments 1 and 2), despite the fact that this antecedent does not occupy a structurally prominent c-commanding position with respect to the reflexive. We further conducted a sentence completion experiment to test whether the findings from the reading experiments could be interpreted as a form of antecedent prediction. We failed to find any evidence that comprehenders generate a strong expectation for coreference to the more recent antecedent inside the embedded clause; although we did find evidence that comprehenders do generally predict that the subject of the embedded clause will be coreferent with one of the two possible antecedents, our results suggested that both nouns were expected to a roughly equal degree.

Overall, these results fail to support the *structure guidance* hypothesis: we observed no evidence that comprehenders preferentially link the fronted reflexives to structurally prominent, c-commanding subjects. Instead, the results are more compatible with the *non-structural search hypothesis*, which holds that in this syntactic environment, syntactic prominence and c-command no longer appear to guide the antecedent retrieval process. In this sense, the present result contrasts with the broader literature on reflexives, which has largely shown that comprehenders prefer to resolve direct object reflexives to the most structurally prominent antecedent in the local syntactic domain (Dillon, 2014; Jäger et al., 2017; Sturt, 2013).

Recency bias and the source of structure sensitivity

The present results raise an interesting question: when, and why, do comprehenders use structural cues to guide the interpretation of a reflexive? One simple and *a priori* plausible answer to this question might be that structurally prominent subjects in a sentence are inherently visible or highly activated during the course of incremental processing, because of their structural prominence, and that this is why comprehenders readily link reflexives to these antecedents in real-time comprehension. However, this simple view predicts that reflexives should always be associated with structurally prominent subjects, regardless of structural context. Because we saw an N2 advantage in both Experiment 1 and Experiment 2 for reflexives in fronted *wh*-predicates, our results

do not support this view. Taken together with previous studies, our results suggest that the retrieval cues recruited to select a reflexive's antecedent are partially a function of the environment the reflexive is found in.

Following others (Cunnings & Sturt, 2014; Jäger et al., 2017; Nicol & Swinney, 1989; Patil et al., 2016; Sturt, 2003, 2013; Xiang et al., 2009), we propose that reflexives in direct object position access their antecedent using a retrieval process that is guided at least in part by structural cues that identify syntactically prominent, local subjects. However, we hasten to add there remains substantial debate over the relative weighting of non-structural cues such as gender and number features in the resolution of reflexive dependencies, as well as whether or not the structural constraints are violated in early reflexive processing (see Jäger et al., 2017 for an in-depth discussion of these issues). We hypothesize that it is possible for the parser to use syntactic cues such as locality and c-command to identify an antecedent for in-situ reflexives because in this sentence context, the parser has access to all the information necessary to apply these constraints; that is, it has constructed a fully connected phrase marker to that point, and that phrase marker encodes structural information necessary to evaluate what antecedents are both c-commanding and structurally local. In contrast, this may not be true for a reflexive in a fronted *wh*-predicate. In this situation, the parser has not yet constructed a full parse of the embedded clause. Because the ultimate syntactic position of fronted *wh*-predicate is temporarily uncertain or unclear, the syntactic structure constructed at the point of encountering the reflexive does not encode the necessary information to evaluate whether potential antecedents are c-commanding or structurally local to the reflexive. If this line of reasoning is correct, then this means that any retrieval cues that make reference to locality or c-command may not be available when the reflexive inside the *wh*-predicate is encountered. Thus, because the reflexive inside the fronted *wh*-predicate has not been integrated with its base position, at this point in incremental processing the parser does not have access to the information necessary to apply structural information like c-command and locality to restrict the antecedent retrieval process, and so it does not.

Without structural information like c-command and locality to guide antecedent retrieval, we saw a recency bias emerge: there was consistent evidence for this in Experiment 1, though the evidence for this preference was weaker in Experiment 2. Assuming that the reflexives inside fronted *wh*-predicates do preferentially associate with the more linearly recent N2, the question arises as to why this should be so. One possibility is that this recency bias reflects the general constraints on the activation of antecedents in memory. The more recent antecedent may be more available due to simple recency or time-based decay (Lewis, Vasishth, & Van Dyke, 2006): since it was processed more recently, it may have a higher resting activation in memory at the point when the reflexive is encountered. However, this simple hypothesis does not take into account the potential impact of the processing events that intervene between N2 and the reflexive. If processing the matrix verb requires a retrieval of the head noun N1, then this could raise the activation level of N1, potentially giving it greater activation than that of N2 (see Dillon, Mishler, Sloggett, & Phillips, 2013, for ACT-R simulations that illustrate the impact of this verb-driven reactivation process on activation). More generally, a given antecedent's activation is a function of how and when it is reactivated over the course of a parse. Given this, this explanation for the N2 effect would be most convincing if it could be demonstrated using an implemented retrieval model that N2 has an activation advantage at the point of encountering the reflexive. An alternative explanation for the recency bias, however, is that the N2 position in our materials was more accessible in memory because it was typically a proper name with unambiguous gender information; the greater clarity of the gender cues may have made this antecedent a better match with the gender features of the reflexive, which could have in turn made it more likely to be retrieved or reactivated when the reflexive is encountered. We are at present unable to

conclusively determine which of these two possibilities best explains the recency bias.

Similarly, it remains to be seen exactly what retrieval cues are involved in the antecedent search for reflexives in fronted *wh*-predicates. The present study demonstrated that structural prominence (i.e., c-command) and locality does not guide the antecedent search in this context, but the search process may still involve other types of cues that are plausibly structural in nature, such as [+subject]. A [+subject] cue would reactivate both subjects (*the mechanic, James*) regardless of their c-command relations with the reflexive, with perhaps an advantage for the more recent and hence more active item (*James*). As such, while a simple recency bias is sufficient to explain the present findings, the findings remain consistent with a model where some structural cues like subjecthood are used. If the [+subject] cue is actually used, it is expected that the parser would not consider the more recent noun if it is in an object position (e.g., *The mechanic that hired James predicted how annoyed with himself...*). We leave this question open for future research.

It is interesting to contrast the present results with the findings on the resolution of cataphoric pronominal dependencies. Several studies present evidence that suggests that comprehenders engage a forward search process when they encounter a pronoun that does not have an antecedent in the preceding context (Aoshima et al., 2009; Cowart & Cairns, 1987; Van Gompel & Liversedge, 2003; Yoshida et al., 2014). Reflexives seem to behave similarly when there is no possible antecedent in the preceding sentence context (Frazier et al., 1996). The present findings contrast interestingly with this body of work in suggesting that given a choice between engaging a forward search to resolve the reflexive (the only strategy that will reliably deliver a grammatical interpretation), and a backwards search through the preceding context, comprehenders preferred the latter strategy. At present it is not clear, however, whether this same preference for backwards search would pertain to other types of pronouns, or whether it is specific to reflexive dependencies. Further research is necessary to determine how general the preference for backwards search over forward search is.

Conclusion

In three experiments, we investigated how comprehenders process reflexives in cataphoric *wh*-predicate fronting constructions. We found that comprehenders initially interpreted these cataphoric reflexives anaphorically, retrieving an antecedent from the preceding context before in place of executing a forward search for the reflexive's antecedent. This pattern suggests that comprehenders prioritize incremental interpretation and prefer to resolve the reflexive's reference in the preceding context instead of engaging a forward search for the grammatical antecedent of the reflexive. Furthermore, we observed that reflexives in fronted *wh*-predicates show a recency bias for their antecedent, rather than the bias towards structurally prominent antecedents that has been observed for anaphoric reflexives in other studies. We proposed that this processing profile arises when comprehenders lack sufficient bottom-up information about the syntactic structure to apply structural cues to guide antecedent retrieval.

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Appendix A. Supplementary material

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jml.2019.04.003>.

References

- Aoshima, S., Phillips, C., & Weinberg, A. (2004). Processing filler-gap dependencies in a head-final language. *Journal of Memory and Language*, 51, 23–54. <https://doi.org/10.1016/j.jml.2004.03.001>.
- Aoshima, S., Yoshida, M., & Phillips, C. (2009). Incremental processing of coreference and binding in Japanese. *Syntax*, 12, 93–134. <https://doi.org/10.1111/j.1467-9612.2009.00123.x>.
- Arnold, J. E. (2010). How speakers refer: The role of accessibility. *Language and Linguistics Compass*, 4, 187–203. <https://doi.org/10.1111/j.1749-818X.2010.00193.x>.
- Arnold, J. E., & Griffin, Z. M. (2007). The effect of additional characters on choice of referring expression: Everyone counts. *Journal of Memory and Language*, 56, 521–536. <https://doi.org/10.1016/j.jml.2006.09.007>.
- Badecker, W., & Straub, K. (2002). The processing role of structural constraints on interpretation of pronouns and anaphors. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, 28, 748–769. <https://doi.org/10.1037/0278-7393.28.4.748>.
- Barr, D. J., Levy, R., Scheepers, C., & Tily, H. J. (2013). Random effects structure for confirmatory hypothesis testing: Keep it maximal. *Journal of Memory and Language*, 68, 255–278. <https://doi.org/10.1016/j.jml.2012.11.001>.
- Barss, A. (1986). *Chains and anaphoric dependence: On reconstruction and its implications* (unpublished doctoral dissertation)Cambridge, MA: Massachusetts Institute of Technology.
- Bates, D., Maechler, M., Bolker, B., & Walker, S. (2015). Fitting linear mixed-effects models using lme4. *Journal of Statistical Software*, 67, 1–48. <https://doi.org/10.18637/jss.v067.i01>.
- Box, G. E. P., & Cox, D. R. (1964). An analysis of transformations. *Journal of the Royal Statistical Society Series B (Methodological)*, 26, 211–252. <https://doi.org/10.1111/j.2517-6161.1964.tb00553.x>.
- Carreiras, M., Garnham, A., Oakhill, J., & Cain, K. (1996). The use of stereotypical gender information in constructing a mental model: Evidence from English and Spanish. *Quarterly Journal of Experimental Psychology: Section A*, 49, 639–663. <https://doi.org/10.1080/027249896392531>.
- Chapman, C. (2018). *Processing LF structure: Evidence from reconstruction for binding* (unpublished doctoral dissertation)McMaster University.
- Chomsky, N. (1981). *Lectures on government and binding: The Pisa lectures*. Berlin: Mouton de Gruyter.
- Chomsky, N. (1986). *Knowledge of language: Its nature, origin, and use*. Connecticut: Greenwood Publishing Group.
- Cowart, W., & Cairns, H. S. (1987). Evidence for an anaphoric mechanism within syntactic processing: Some reference relations defy semantic and pragmatic constraints. *Memory & Cognition*, 15, 318–331. <https://doi.org/10.3758/BF03197034>.
- Cummings, I., & Felsner, C. (2013). The role of working memory in the processing of reflexives. *Language and Cognitive Processes*, 28, 188–219. <https://doi.org/10.1080/01690965.2010.548391>.
- Cummings, I., & Sturt, P. (2014). Coargumenthood and the processing of reflexives. *Journal of Memory and Language*, 75, 117–139. <https://doi.org/10.1016/j.jml.2014.05.006>.
- Dillon, B. (2014). Syntactic memory in the comprehension of reflexive dependencies: An overview. *Language and Linguistics Compass*, 8, 171–187. <https://doi.org/10.1111/lnc3.12075>.
- Dillon, B., Mishler, A., Sloggett, S., & Phillips, C. (2013). Contrasting intrusion profiles for agreement and anaphora: Experimental and modeling evidence. *Journal of Memory and Language*, 69(2), 85–103.
- Federmeier, K. D., & Kutas, M. (1999). A rose by any other name: Long-term memory structure and sentence processing. *Journal of Memory and Language*, 41, 469–495. <https://doi.org/10.1006/jmla.1999.2660>.
- Foraker, S., & McElree, B. (2007). The role of prominence in pronoun resolution: Active versus passive representations. *Journal of Memory and Language*, 56, 357–383. <https://doi.org/10.1016/j.jml.2006.07.004>.
- Fox, D., & Nissenbaum, J. (2004). Condition A and scope reconstruction. *Linguistic Inquiry*, 35, 475–485. <https://doi.org/10.1162/ling.2004.35.3.475>.
- Frazier, L., Plunkett, B., & Clifton, C. (1996). Reconstruction and scope. *University of Massachusetts Occasional Papers in Linguistics*, 19, 239–260.
- Gelman, A., & Hill, J. (2007). *Data analysis using regression and multilevel hierarchical models*, Vol. 1. New York: Cambridge University Press.
- Heycock, C. (1995). Asymmetries in reconstruction. *Linguistic Inquiry*, 26, 547–570.
- Huang, C. T. J. (1993). Reconstruction and the structure of VP: Some theoretical consequences. *Linguistic Inquiry*, 24, 103–138.
- Jäger, L. A., Engelmann, F., & Vasishth, S. (2017). Similarity-based interference in sentence comprehension: Literature review and Bayesian meta-analysis. *Journal of Memory and Language*, 94, 316–339. <https://doi.org/10.1016/j.jml.2017.01.004>.
- Kaiser, E., Runner, J. T., Sussman, R. S., & Tanenhaus, M. K. (2009). Structural and semantic constraints on the resolution of pronouns and reflexives. *Cognition*, 112, 55–80. <https://doi.org/10.1016/j.cognition.2009.03.010>.
- Kazanina, N., Lau, E. F., Lieberman, M., Yoshida, M., & Phillips, C. (2007). The effect of syntactic constraints on the processing of backwards anaphora. *Journal of Memory and Language*, 56, 384–409. <https://doi.org/10.1016/j.jml.2006.09.003>.
- Kennison, S., & Trofe, J. L. (2003). Comprehending pronouns: A role for word-specific gender stereotype information. *Journal of Psycholinguistic Research*, 32, 355–378. <https://doi.org/10.1023/A:1023599719948>.
- Lewis, R. L., Vasishth, S., & Van Dyke, J. A. (2006). Computational principles of working memory in sentence comprehension. *Trends in Cognitive Sciences*, 10, 447–454. <https://doi.org/10.1016/j.tics.2006.08.007>.
- Nicol, J., & Swinney, D. (1989). The role of structure in coreference assignment during sentence comprehension. *Journal of Psycholinguistic Research*, 18, 5–19. <https://doi.org/10.1007/BF01069043>.
- Omaki, A. (2010). *Commitment and flexibility in the developing parser* (unpublished doctoral dissertation)College Park, MA: University of Maryland.
- Parker, D., & Phillips, C. (2017). Reflexive attraction in comprehension is selective. *Journal of Memory and Language*, 94, 272–290. <https://doi.org/10.1016/j.jml.2017.01.002>.
- Patil, U., Vasishth, S., & Lewis, R. L. (2016). Retrieval interference in syntactic processing: The case of reflexive binding in English. *Frontiers in Psychology*, 7, 1–18. <https://doi.org/10.3389/fpsyg.2016.00329>.
- R Development Core Team (2015). *R: A Language and Environment for Statistical Computing*. Vienna: R Foundation for Statistical Computing. <http://www.R-project.org/>.
- Rayner, K. (1998). Eye movements in reading and information processing: 20 years of research. *Psychological Bulletin*, 124, 372–422. <https://doi.org/10.1037/0033-2909.124.3.372>.
- Rayner, K., & Pollatsek, A. (2006). *Eye-movement control in reading*. New York, NY: Academic Press.
- Runner, J. T., Sussman, R. S., & Tanenhaus, M. K. (2003). Assignment of reference to reflexives and pronouns in picture noun phrases: Evidence from eye movements. *Cognition*, 89, B1–B13. [https://doi.org/10.1016/S0010-0277\(03\)00065-9](https://doi.org/10.1016/S0010-0277(03)00065-9).
- Runner, J. T., Sussman, R. S., & Tanenhaus, M. K. (2006). Processing reflexives and pronouns in picture noun phrase. *Cognitive Science*, 30, 193–241. <https://doi.org/10.1207/s15516709cog0000.58>.
- Schnoebelen, T., & Kuperman, V. (2010). Using Amazon Mechanical Turk for linguistic research. *Psihologija*, 43, 441–464. <https://doi.org/10.2298/PSI1004441S>.
- Sloggett, S. (2017). *When errors aren't: How comprehenders selectively violate binding theory* (unpublished doctoral dissertation)Amherst, MA: University of Massachusetts Amherst.
- Sportiche, D. (2006). Reconstruction, binding, and scope. In M. Everaert, & H. Riemsdijk (Vol. Eds.), *The Blackwell companion to syntax: Vol. 4*, (pp. 35–93). Oxford: Blackwell.
- Staub, A., Grant, M., Astheimer, L., & Cohen, A. (2015). The influence of cloze probability and item constraint on cloze task response time. *Journal of Memory and Language*, 82, 1–17. <https://doi.org/10.1016/j.jml.2015.02.004>.
- Staub, A., & Rayner, K. (2007). Eye movements and on-line comprehension processes. In G. Gaskell (Ed.), *The Oxford handbook of psycholinguistics* (pp. 327–342). Oxford: Oxford University Press.
- Sturt, P. (2003). The time-course of the application of binding constraints in reference resolution. *Journal of Memory and Language*, 48, 542–562. [https://doi.org/10.1016/S0749-596X\(02\)00536-3](https://doi.org/10.1016/S0749-596X(02)00536-3).
- Sturt, P. (2013). Syntactic constraints on referential processing. In R. P. Van Gompel (Ed.), *Sentence processing* (pp. 136–159). Hoboken: Taylor and Francis.
- Takano, Y. (1995). Predicate fronting and internal subjects. *Linguistic Inquiry*, 26, 327–340.
- Van Gompel, R. P., & Liversedge, S. P. (2003). The influence of morphological information on cataphoric pronoun assignment. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, 29, 128–139. <https://doi.org/10.1037/0278-7393.29.1.128>.
- von der Malsburg, T., & Angele, B. (2017). False positives and other statistical errors in standard analyses of eye movements in reading. *Journal of memory and language*, 94, 119–133.
- Xiang, M., Dillon, B., & Phillips, C. (2009). Illusory licensing effects across dependency types: ERP evidence. *Brain and Language*, 108, 40–55. <https://doi.org/10.1016/j.bandl.2008.10.002>.
- Yoshida, M., Kazanina, N., Pablos, L., & Sturt, P. (2014). On the origin of islands. *Language, Cognition and Neuroscience*, 29, 761–770. <https://doi.org/10.1080/01690965.2013.788196>.